

AYE
TECH HUB

• PLC PROGRAMMING • LESSON 05

Timer and Counter Instructions

TON · TOF · RTO · CTU · CTD · CTUD

Lesson 05 of 30 | PLC Programming Program

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- TON, TOF, and RTO timers — how each works and when to use each type
- Timing diagrams: visualising input, elapsed time, and output behaviour
- CTU, CTD, and CTUD counters — count up, count down, and bidirectional
- Complete ladder logic rungs for timers and counters with all parameters
- Real-world applications: conveyor delays, batch counting, alarm timing

Timers — Controlling Sequences with Time

Timers are among the most used instructions in any PLC program. They introduce time delays between events — starting a motor 5 seconds after a button press, holding a valve open for 30 seconds, or sounding an alarm if a condition persists for more than 10 seconds. **Almost every industrial process involves time-based sequencing, and timers are the tool that implements it.**

Figure 1 — Timer Timing Diagrams: TON / TOF / RTO (IN = input, Q = output, PT = preset time)



Timer Type	Full Name	Behaviour	Typical Use Case
TON	Timer On-Delay	Output Q goes TRUE after input IN has been TRUE continuously for preset time PT. If IN goes FALSE before PT, elapsed time resets to zero.	Motor start delay after button press. Conveyor delay before part enters station. Alarm activation after fault condition persists.
TOF	Timer Off-Delay	Output Q goes FALSE after input IN has been FALSE continuously for preset time PT. Q is TRUE immediately when IN is TRUE.	Fan continues running for 60 seconds after machine stops. Lights stay on for 30 seconds after presence sensor clears.
RTO	Retentive Timer (Accumulating)	Accumulates elapsed time ET across multiple input pulses. ET does not reset when IN goes FALSE — only resets with explicit RES instruction.	Tracking total run hours of a motor. Measuring cumulative machine operation time for maintenance scheduling.
TP	Pulse Timer	Generates a fixed-duration output pulse regardless of how long input stays TRUE. Output stays high for exactly PT, then drops.	Triggering a solenoid for a fixed 200ms stroke. Creating a fixed-width enable signal for downstream logic.

Timer Parameter	Data Type	Description
IN (Input)	BOOL	Enable input — when TRUE, timer starts accumulating elapsed time
PT (Preset)	TIME	The target duration. Format: T#5s (5 sec), T#500ms, T#2m30s
ET (Elapsed)	TIME	Current elapsed time — readable tag showing how long timer has been running
Q (Output)	BOOL	Timer done bit — TRUE when ET ≥ PT (for TON/TP) or timing off (TOF)
R (Reset)	BOOL	Reset input (RTO/RES only) — resets ET to zero when TRUE

Time format in IEC 61131-3 PLC programming: T#5s = 5 seconds. T#500ms = 500 milliseconds. T#2m = 2 minutes. T#1h30m = 1 hour 30 minutes. T#0d1h2m3s4ms = full format. In Siemens TIA Portal: S5T#5S (older S5 format) or T#5s (IEC format). Always confirm which time format your PLC platform uses.

Counters — Tracking Events and Quantities

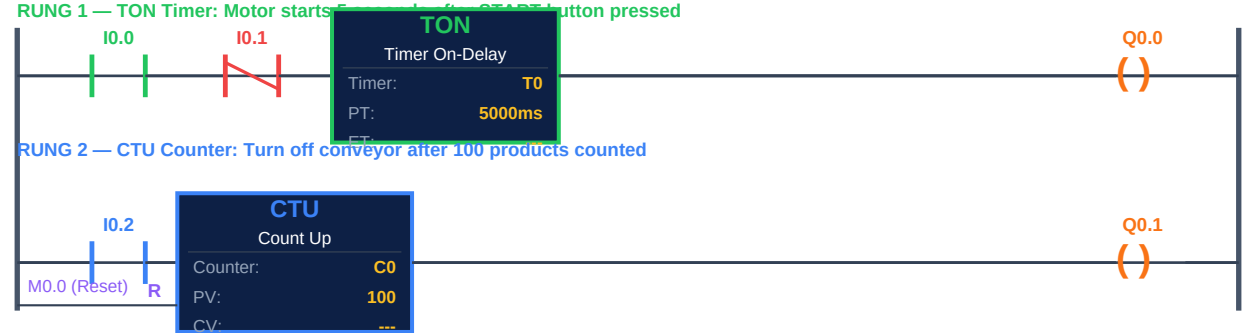
Counters track how many times an event occurs — how many parts have passed a sensor, how many cycles a machine has completed, how many errors have occurred. When the count reaches a preset value, the counter output triggers the next action.

Figure 3 — Counter Types: CTU / CTD / CTUD — Parameters and Behaviour

CTU Count Up	CTD Count Down	CTUD Up/Down
CU: count pulse input R: reset input (→ CV=0) PV: preset value CV: current value Q: ON when $CV \geq PV$ 2 → ... → PV Q turns ON at PV	CD: count pulse input LD: load input (→ CV=PV) PV: preset value CV: current value Q: ON when $CV \leq 0$ 1 → ... → 1 → 0 Q turns ON at zero	CU: count up input CD: count down input R: reset; LD: load PV: preset QU: $CV \geq PV$, QD: $CV \leq 0$ Bidirectional QU and QD outputs

Counter Type	Full Name	Behaviour	Typical Use Case
CTU	Count Up	Increments CV by 1 on each rising edge (0→1 transition) of CU input. Q output goes TRUE when $CV \geq PV$. R input resets CV to 0.	Count products on a conveyor. Count machine cycles before maintenance. Count operator button presses.
CTD	Count Down	Decrements CV by 1 on each rising edge of CD input. Q output goes TRUE when $CV \leq 0$. LD (Load) input resets CV back to PV.	Count remaining items in a batch. Count down remaining seconds on a work timer.
CTUD	Count Up/Down	Bidirectional counter with both CU (count up) and CD (count down) inputs. QU goes TRUE when $CV \geq PV$. QD goes TRUE when $CV \leq 0$. R resets to 0; LD loads PV.	Tracking items entering and leaving a buffer zone. People counter at an entrance (in/out).

Figure 2 — Ladder Logic: TON Timer Rung (Rung 1) and CTU Counter Rung (Rung 2)



Counter edge detection — critical rule: Counters increment on the rising edge (FALSE→TRUE transition) of the count input, NOT on the level. If the sensor input stays TRUE for multiple scan cycles, the counter only increments ONCE per TRUE pulse. If your sensor gives a continuous signal, add a one-shot (P contact or R_TRIG block) upstream of the counter to convert level to edge before counting.

Real-World Applications and Programming Tips

Application	Timer/Counter Used	PLC Logic Summary
Conveyor start delay	TON (T#3s)	Start button (I0.0) enables TON. After 3 seconds, TON.Q energises conveyor motor coil (Q0.0). Stop button NC contact in series.
Alarm on sustained fault	TON (T#10s)	Fault sensor (I0.1) drives TON. If fault persists > 10s, TON.Q activates alarm output (Q0.1). Fault clearing resets timer automatically.
Batch count — stop after 50 parts	CTU (PV=50)	Photoelectric sensor (I0.2) increments CTU on each rising edge. When CV=50: CTU.Q de-energises conveyor. Operator presses Reset to restart.
Motor cooldown after stop	TOF (T#60s)	Motor run signal drives TOF. Motor stops → TOF starts. Cooling fan output (Q0.2) stays ON for 60 seconds after motor stops.
Total run-hours tracking	RTO accumulates while motor runs	Motor coil bit drives RTO IN. RTO.ET stored to data register each scan. Maintenance alarm when ET reaches T#500h. Reset only on maintenance completion.
People counter (entry/exit)	CTUD (PV=max capacity)	Entry sensor drives CU input. Exit sensor drives CD input. CV = current occupancy. QU turns on "FULL" sign when CV ≥ PV.
Pulse output for solenoid	TP (T#200ms)	Trigger bit (M0.0) drives TP. TP.Q energises solenoid for exactly 200ms regardless of how long M0.0 stays TRUE.

Platform	Timer Block Name	Counter Block Name	Time Format Example
Siemens S7 (TIA Portal IEC)	IEC_TIMER (TON, TOF, TP, TONR)	IEC_COUNTER (CTU, CTD, CTUD)	T#5s or T#500ms
Siemens S7 (Classic / S5)	S_ODT (on-delay), S_OFFDT (off-delay)	S_CU (count up), S_CD (count down)	S5T#5S (S5 time format)
Allen-Bradley / Rockwell	TON, TOF, RTO (IEC-compatible)	CTU, CTD, RES (reset)	Preset in milliseconds: 5000 = 5s
Mitsubishi MELSEC	OUT T (timer coil instruction)	OUT C (counter coil instruction)	K50 = 50 × 100ms = 5 seconds
Omron / Schneider (IEC 61131-3)	TON, TOF, TP, TONR	CTU, CTD, CTUD	T#5s (standard IEC format)

What comes next — PLC-006: Data Handling and Move Instructions — MOV, MOVE, FILL, and data type conversions. How to read sensor values into data registers, compare values, move data between tags, and use indirect addressing. The foundation for any PLC program that processes real data beyond simple ON/OFF logic.